GyuYeop Do

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Education

Seoul National University March 2020 - July 2026
B.S. Electrical & Computer Engineering Seoul, Republic of Korea

- Includes 1.5 years of mandatory military service as a Korean Augmentation to the United States Army (KATUSA) soldier
- Relevant Coursework: Deep Learning, 3D Computer Vision, Computer Systems Programming, Linear Algebra

Research Experience

Seoul National University - COSS Program December 2025 - Present
SNU COSS Group (Advisor: Nam-Joon Kim) Seoul, Republic of Korea

- Conducting research on open-vocabulary segmentation for edge devices.

Seoul National University - Electrical and Computer Engineering September 2025 - December 2025
Machine Perception and Reasoning Lab (Advisor: Jonghyun Choi) Seoul, Republic of Korea

- Undergraduate Thesis: Enhancing action prediction in Vision-Language-Action (VLA) models with 3D foundation models.

Seoul National University - Computer Science December 2024 - July 2025
Visual & Geometric Intelligence Lab (Advisor: Jaesik Park) Seoul, Republic of Korea

- Implemented key modules of an RGB-only grasp prediction pipeline based on normal-field estimation to improve robotic grasping of transparent objects.
- Built an RGB-only grasp prediction pipeline leveraging feed-forward 3D feature inference and surface normal prediction for objects challenging for conventional 3D reconstruction methods.
- Developed a method for reconstructing signed distance functions (SDFs) of dynamic rigid objects from images captured during motion.

Publications

- **GyuYeop Do**, Jiho Ryou, Jinwoo Jeon, Ahn Jin Yeong, Sunwoong Kim, Hyuk-Jae Lee, Hyungon Ryu, and Nam-Joon Kim. “CoMoE: Collaborative Expert Aggregation for Sparse Mixture-of-Experts.” ACL ARR 2026 May Submission, under review, 2026.
- Jinwoo Jeon, **GyuYeop Do**, Yubin Lim, Nam-Joon Kim, Hyun Gon Ryu, Hyuk-Jae Lee, and Byung-Jun Lee. “Efficient Quantization-Aware Distillation with Cross-Modal Alignment for Edge Vision–Language Models.” ECCV 2026 Conference Submission, under review, 2026.

Educational Publications

- 2026 Gradient Korean SAT (CSAT) Physics I Prep Test. Team GRAVITY, 2025.
- 2026 Gravity Korean SAT (CSAT) Physics I Prep Test. Team GRAVITY, 2025.
- 2025 Gravity Korean SAT (CSAT) Physics I Prep Test. Team GRAVITY, 2024.
- 2024 Gravity Korean SAT (CSAT) Physics I Prep Test. Team GRAVITY, 2023.
- 2023 Gravity Korean SAT (CSAT) Physics I Prep Test. Team GRAVITY, 2022.

Experience

Team Gravity March 2020 - Present
Founder Seoul, Republic of Korea

- Founded and led an educational content business producing Korean SAT Physics study materials, publishing 6+ commercially sold prep books available in major bookstores, and selling content to one of South Korea's largest education enterprises.
- Managed a team of ~20 physics content developers and 5 illustrators over 6 years, overseeing content development, quality control, and production for high-difficulty physics problem sets.

Undergraduate Deep Learning Club January 2025 - Present
Club President Seoul, Republic of Korea

- Led an undergraduate deep learning club across three cohorts, each focused on an in-depth study track in Reinforcement Learning, Computer Vision, and NLP.
- Facilitated weekly paper readings, discussions, and project-based learning, with each cohort completing a final project.

AWS & Seoul National University - Data Venture Hackathon July 2025 - Present
Team Lead Seoul, Republic of Korea

- Team lead in the AWS Korea & SNU Data Venture Hackathon, developing a platform for franchise owners.
- Contributed to the majority of the codebase, implementing a multi-agent AI system, a backend built with Spring Boot, and a frontend developed using Next.js.
- Built a robust AI system with multiple agents and RAG to answer franchise owners' questions based on the brand's info and Korean legal documents.

POSTECH Big Data Challenge - Solar Irradiance Estimation Competition November 2025 - December 2025
Team Lead Republic of Korea

- Conducted tabular data analysis to model solar irradiance using LightGBM and XGBoost with an ensemble approach.
- Conducted extensive feature engineering and data visualization to improve accuracy.

Qualcomm & Microsoft - Edge AI Hackathon July 2025 - August 2025
Team Lead Seoul, Republic of Korea

- Led development of a fully offline, real-time generative storytelling app on a Snapdragon-based Windows laptop, with a PySide desktop interface.

- Optimized Llama 3.2 3B for real-time on-device text generation by compiling it with the Qualcomm QNN SDK for NPU acceleration.
- Integrated CLIP and Stable Diffusion, compiling and running Stable Diffusion on the NPU via ONNX Runtime for efficient edge inference.

AWS & Seoul National University - Solving a Social Problem Hackathon January 2025 - Present
Main Developer Seoul, Republic of Korea

- Developed an Android application using Jetpack Compose integrated with AWS services.
- Implemented an AI-driven system using RAG and speech-to-text (STT) to automatically warn users of potential scams or fraudulent content immediately after messages or phone calls.

Republic of Korea Army - Korean Augmentation to the United States Army November 2022 - May 2024
Pyeongtaek, Republic of Korea

- Completed 1.5 years of mandatory military service as Korean Augmentation to the United States Army (KATUSA), serving in the 1st Signal Brigade, U.S. Army.

Teaching

Sidae Injae Academy - Korean SAT Physics Prep January 2019 - July 2019
TA for Instructor Song-Do Kim Seoul, Republic of Korea

- Supported Song-Do Kim, a recognized instructor with 1,000+ students in large-scale offline lectures.
- Developed core instructional materials, including problem sets, quizzes, assignments used across classes.

Sidae Injae Academy - Advanced High School Mathematics January 2019 - July 2019
TA for Instructor Young-Seok Park Seoul, Republic of Korea

- Assisted Young-Seok Park in teaching classes of 100+ students by providing one-on-one explanations for advanced problem-solving and creating textbooks and assessment materials.

Awards

Army Commendation Medal (U.S. Army) July 2024
Awarded by HQ Eighth Army during service in the 1st Signal Brigade, U.S. Army
Approved by Brigadier General Roderick F. Laughman (USA)
Recognized for exceptional service and contribution during military duty in coordination with the U.S. Army.

Exemplary KATUSA Commendation February 2024
Awarded by 2nd Area Commander, Republic of Korea Army (ROKA) Support Group Lieutenant Colonel Kim, Ki-Hyun (ROKA)
Recognized for exceptional professionalism, dedication, and exemplary military conduct. Demonstrated leadership, discipline, and performance that set a high standard within the KATUSA (Korean Augmentation to the United States Army) program.

AWS & Seoul National University - Data Venture Hackathon October 2025
Top 6 Finalist.

Pohang University of Science and Technology - Big Data Challenge December 2025
Bronze award. Ranked top 7 of 140 participants.

Qualcomm & Microsoft - Edge AI Hackathon August 2025
4th Place Award.